

CO3 - AT1- Analytical Problem solving

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QUESTIONS

Q1
CO3 - AT1 - Analytical Problem Solving
20 marks

Q2
CO3 - AT1 - Analytical Problem Solving
20 marks

Q3
CO3 - AT1 - Analytical Problem solving
20 marks

Q4
CO3 - AT1 - Analytical Problem Solving
20 marks

Q5
CO3 - AT1 - Analytical Problem Solving
20 marks

5/5 answered

Q1 CO3 - AT1 - Analytical Problem Solving

20 marks

1. Given an $N \times N$ matrix, Write a C Program to determine whether it is:
- a) Symmetric
 - b) Upper triangular
 - c) Lower triangular
 - d) Diagonal matrix
 - e) Identity matrix
 - f) Display all applicable classifications.

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int N, i, j;
5     int symmetric = 1, upper = 1, lower = 1, diagonal = 1, identity = 1;
6     int a[50][50];
7
8     printf("Enter N (<= 50): ");
9     scanf("%d", &N);
10
11     printf("Enter the elements of %d x %d matrix:\n", N, N);
12     for (i = 0; i < N; i++) {
13         for (j = 0; j < N; j++) {
14             scanf("%d", &a[i][j]);
15         }
16     }
17 }
```

Input

3
1 2 3
2 3 4
3 4 5

Output



CO3 - AT2 - Class Test

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QUESTIONS

Q1
CO3 - AT2 - Class Test
20 marks

Q2
CO3 - AT2 - Class Test
20 marks

Q3
CO3 - AT2 - Class Test
20 marks

Q4
CO3 - AT2 - Class Test
20 marks

Q5
CO3 - AT2 - Class Test
20 marks

5/5 answered

Q1 CO3 - AT2 - Class Test

20 marks

1. Write a program to count the frequency of each element in an array.

Sample Input: 2 2 1 5 1 3 3 (7 elements)

Sample Output: 1: 2 times, 2: 2 times, 3: 2 times, 5: 1 time

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     int a[n];
8     for (int i = 0; i < n; i++) scanf("%d", &a[i]);
9
10    int done[n];
11    for (int i = 0; i < n; i++) done[i] = 0;
12
13    for (int i = 0; i < n; i++) {
14        if (done[i]) continue;
15
16        int count = 1;
17        done[i] = 1;
18
19        for (int j = i + 1; j < n; j++) {
20            if (!done[j] && a[j] == a[i]) {
```

Input

2 2 1 5 1 3 3

Output



CO3 - AT3 - Mock Interview

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QUESTIONS

Q1

CO3 - AT3 - MOCK INTERVIEW
- ORAL ANSWERS

100 marks

0/1 answered

Q1 CO3 - AT3 - MOCK INTERVIEW - ORAL ANSWERS

100 marks

- 1.What is the difference between an array and a variable?
2. What is meant by the base address of an array?
- 3.What happens if an array size is not specified during initialization?
4. How are 2D arrays stored in memory?
5. Give one real-life application of multidimensional arrays.
- 6.Why do arrays start from index 0 instead of 1?
- 7.Which engineering fields commonly use multidimensional arrays?
- 8.Can a multidimensional array have more than three dimensions?
- 9.Why are images stored as multidimensional arrays?
10. What is row-major order?
11. Does C use row-major or column-major order?
12. Why can't arrays return multiple values?
13. What happens when array bounds are exceeded?
14. Why are strings stored as character arrays?
15. Explain the difference between `char str[]` and `char *str`.
16. Can two-dimensional arrays be passed to functions?
17. When would a linked list outperform an array?
18. Differentiate `gets()`, `fgets()`, `scanf()`, and `getchar()`.

Your Code

1

Input

stdin input